First-order equations

The method of characteristics

- Focus on the case of two independent variables, $u = u(x, y)$.
- Along a characteristic curve (or characteristic) in the (x, y) plane, the PDE can be integrated directly to give the solution u.
- To use this method, we need the region on which we are solving the PDE to be covered by characteristic curves.

Theory

- We focus on the case

$$A(x, y)\frac{\partial u}{\partial x} + B(x, y)\frac{\partial u}{\partial y} = F(x, y)$$

and consider the family of curves (the characteristics) defined by

$$\frac{dx}{A} = \frac{dy}{B} \qquad \text{or} \qquad \frac{dy}{dx} = \frac{B}{A}$$

- The element of arc length (squared) along one of these curves is given by

$$ds^2 = dx^2 + dy^2$$

- Since $dx = (A/B)\,dy$, we can also write

$$ds^2 = \frac{A^2}{B^2}dy^2 + dy^2 = \frac{1}{B^2}(A^2 + B^2)\,dy^2$$

and $$ds = \frac{1}{B}\sqrt{A^2 + B^2}\,dy \qquad (1)$$

- Similarly, $$ds = \frac{1}{A}\sqrt{A^2 + B^2}\,dx \qquad (2)$$

- If we multiply the original PDE by ds, we find

$$A\,ds\frac{\partial u}{\partial x} + B\,ds\frac{\partial u}{\partial y} = F\,ds$$

or, using (1) and (2)

$$A\frac{\sqrt{A^2 + B^2}}{A}dx\frac{\partial u}{\partial x} + B\frac{\sqrt{A^2 + B^2}}{B}dy\frac{\partial u}{\partial y} = F\,ds$$

$$\sqrt{A^2 + B^2}\left\{\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy\right\} = F\,ds$$

- The term in braces is the total differential of u, du, so

$$\frac{du}{F} = \frac{ds}{\sqrt{A^2 + B^2}}$$

- Using 1 and 2 again, we also have that

$$\boxed{\frac{du}{F} = \frac{dx}{A} = \frac{dy}{B}}$$

from which the solution u can be found by integration along the characteristics.